

Course

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4.1. Practice Lab Assignment

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4.1.1. Pandas - series creation and manipulation

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases

seriesMa...

1 import pandas as pd

2

3 # Take inputs from the user to create a list of numbers

4 numbers = list(map(int, input().split()))

5

6 # Create a Pandas series from the list of numbers

7 s=pd.Series(numbers)

8 # Grouping by even and odd numbers and calculating the mean

9 grouped = s.groupby(s % 2==0).mean()

10

11 # Display the mean of even and odd numbers with labels

12 grouped.index = ['Even' if is_even else 'Odd' for is_even in

13 grouped.index]

14 print("Mean of even and odd numbers:")

15 print(grouped)

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4.1.2. Dictionary to dataframe

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column Gender with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

Remove the specified column.

datafram...

1 # import pandas as pd

2

3 ## Provided dictionary of lists

4 # data = {

5 # 'Name': ['Alice', 'Bob', 'Charlie'],

6 # 'Age': [25, 30, 35],

7 # }

8

9 ## Convert the dictionary to a DataFrame

10 df = pd.DataFrame(data)

11

12 ## Display the original DataFrame

13 # print("Original DataFrame:")

14 # print(df)

15

16 ## Adding a new row

17

18

19 ## Display the DataFrame after adding a new row

20 # print("After adding a row:\n",df)

21

22 ## Modifying a row

23

24

25 ## Display the DataFrame after modifying a row

26 # print("After modifying a row:")

27 # print(df)

28

29 ## Deleting a row

30

31 ## Display the DataFrame after deleting a row

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4.1.3. Student Information

08:56

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

• Display the first five rows of the data frame.

• Calculate the average age of the students (limit the average age up to 2 decimal places).

• Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:
Refer to the displayed test cases for better understanding.

Sample Test Cases

studentin...

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1 # import pandas as pd

2

3 ## Read the text file into a DataFrame

4 # file = input()

5 # data = pd.read_csv(file, sep="\s+", header=None, names=

6 ["Name", "Age", "Grade"])

7

8 # write your code here..

9

10 import pandas as pd

11

12 # Read the text file into a DataFrame

13 file = input()

14 data = pd.read_csv(file, sep="\s+", header=None, names=

15 ["Name", "Age", "Grade"])

16

17 # write your code here..

18

19 print("First five rows:")

20 print(data.head())

21

22 avg_age = round(data["Age"].mean(),2)

23 print(f"Average age: {avg_age}")

24

25 print("Students with a grade up to B")

26

27 filtered_student = data[data["Grade"] <= "B"]

28 print(filtered_student)

29

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5. Practical 5

4.2.1. Month with the Highest Total Sales

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	6	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

1 import pandas as pd

2

3 # Prompt the user for the file name

4 file_name = input()

5

6 # Load the data

7 df = pd.read_csv(file_name)

8

9

10 # Find the month with the highest total sales

11 df["Date"] = pd.to_datetime(df["Date"])

12 df["Sales"] = df["Quantity"] * df["Price"]

13 df["Month"] = df["Date"].dt.to_period("M").astype(str)

14

15 monthly_sales = df.groupby("Month")["Sales"].sum()

16

17 # Find the month with the highest total sales

18 best_month = monthly_sales.idxmax()

19 highest_sales = monthly_sales.max()

20

21 print(f"Best month: {best_month}")

22 print(f"Total sales: \${highest_sales:.2f}")

23

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4.2.2. Best Selling Product

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	5	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9 ps=df.groupby('Product')['Quantity'].sum()
10 # Find the product with the highest total quantity sold
11 best_product = ps.idxmax()
12 highest_quantity = ps.max()
13
14 # Display the result
15 print(f"Best selling product: {best_product}")
16 print(f"Total quantity sold: {highest_quantity}")
17
```

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- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Date	Product	Quantity	Price	City
2025-01-01	Product A	5,20	New York	
2025-01-02	Product B	3,15	Los Angeles	
2025-01-02	Product A	7,20	New York	
2025-01-02	Product C	4,38	Chicago	
2025-01-03	Product B	2,15	Chicago	
2025-01-03	Product A	8,20	Los Angeles	
2025-01-04	Product C	6,30	New York	
2025-01-04	Product B	5,15	Los Angeles	
2025-01-05	Product A	3,20	Chicago	
2025-01-05	Product C	10,30	Los Angeles	

Sample Test Cases

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9 # write the code..
10 cs=df.groupby('City')['Quantity'].sum()
11 best_city=cs.idxmax()
12 # Display the result
13 print(f'City sold the most products: {best_city}")
14
```

Sample Data:

Explanation:

Now, let's count how often the pairs of products appear together:

- **Product A and Product B** (2 times)

Terminal Test cases

4.2.5. Titanic Dataset Analysis and Data Cleaning

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

- Display the first 5 rows of the dataset.
- Display the last 5 rows of the dataset.
- Get the shape of the dataset (number of rows and columns).
- Get a summary of the dataset (using `.info()`).
- Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
- Check for missing values and display the count of missing values for each column.
- Fill missing values in the 'Age' column with the median age.
- Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
- Drop the 'Cabin' column due to many missing values.
- Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cunings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17509,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
```

titanicDat...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # 1. Display the first 5 rows of the dataset
8 print(data.head())
9
10 # 2. Display the last 5 rows of the dataset
11 print(data.tail())
12
13 # 3. Get the shape of the dataset (number of rows and columns)
14 print(data.shape)
15
16 # 4. Get a summary of the dataset
17 print(data.info())
18
19 # 5. Get basic statistics of the dataset
20 print(data.describe())
21
22 # 6. Check for missing values and display the count
23 print(data.isnull().sum())
24
25 # 7. Fill missing values in the 'Age' column with the median age
26
27 data['Age'].fillna(data['Age'].median(), inplace=True)
28
29 # 8. Fill missing values in the 'Embarked' column with the most frequent value
30 (mode)
```

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4.2.8. Titanic Dataset Analysis and Data Cleaning - 2

00:17

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
- Convert the 'Sex' column to numeric values (male: 0, female: 1).
- One-hot encode the 'Embarked' column, dropping the first category.
- Get the mean age of passengers.
- Get the median fare of passengers.
- Get the number of passengers by class.
- Get the number of passengers by gender.
- Get the number of passengers by survival status.
- Calculate the survival rate of passengers.
- Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,1,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Haran, Mr. James",male,,0,0,310077,0.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.6625,E46,S
8,0,3,"Paisson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
```

titanicDat...

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Debugger

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data['FamilySize'] = data['SibSp'] + data['Parch']
7
8 # 1. Create a new column 'IsAlone' (1 if alone, 0 otherwise)
9 data['IsAlone'] = np.where(data['FamilySize'] == 0, 1, 0)
10
11 # 2. Convert 'Sex' to numeric (male: 0, female: 1)
12 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
13
14 # 3. One-hot encode the 'Embarked' column
15 data = pd.get_dummies(data, columns=['Embarked'])
16
17 # 4. Get the mean age of passengers
18 mean_age = data['Age'].mean()
19 print(mean_age)
20
21 # 5. Get the median fare of passengers
22 median_fare = data['Fare'].median()
23 print(median_fare)
24
25 # 6. Get the number of passengers by class
26 print(data['Pclass'].value_counts())
27
28 # 7. Get the number of passengers by gender
29 print(data['Sex'].value_counts())
30
31 # 8. Get the number of passengers by survival status
```

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00:23 🔊 🌙 📝 🔗 —

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

[illegible]

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,Sibsp,Parat,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 2171,7.25,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3101282,7.92,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,S

6,0,3,"Moran, Mr. James",male,0,0,330807,8.4503,C

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,54.8625,E46,S

```

1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data['FamilySize'] = data['SibSp'] + data['Parch']
7 data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
8 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
9
10 print(data.groupby('Pclass') ['Survived'].mean())
11
12 #2. Calculate the survival rate by embarked location (Embarked_S)
13
14 print(data.groupby('Embarked_S') ['Survived'].mean())
15
16 #3. Calculate the survival rate by family size
17
18 print(data.groupby('FamilySize') ['Survived'].mean())
19
20 #4. Calculate the survival rate by being alone
21
22 print(data.groupby('IsAlone') ['Survived'].mean())
23
24 #5. Get the average fare by class
25
26 print(data.groupby('Pclass') ['Fare'].mean())
27
28 #6. Get the average age by class
29
30 print(data.groupby('Pclass') ['Age'].mean())
31

```

4.2.8. Titanic Dataset Analysis and Data Cleaning - 4

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,S
6,0,3,"Moran, Mr. Jones",male,,0,0,330877,8.4503,,0
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
```

titanicDat...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
7
8
9 # 1. Get the number of survivors by gender
10 survivors_by_gender = data[data['Survived'] == 1]['Sex'].value_counts()
11 print(survivors_by_gender)
12
13 # 2. Get the number of non-survivors by gender
14 non_survivors_by_gender = data[data['Survived'] == 0]['Sex'].value_counts()
15 print(non_survivors_by_gender)
16
17 # 3. Get the number of survivors by embarked location (Embarked_S)
18 survivors_by_embarked_s = data[data['Survived'] == 1]['Embarked_S'].value_counts()
19 print(survivors_by_embarked_s)
20
21 # 4. Get the number of non-survivors by embarked location (Embarked_S)
22 non_survivors_by_embarked_s = data[data['Survived'] == 0]
23 ['Embarked_S'].value_counts()
24 print(non_survivors_by_embarked_s)
25
26 # 5. Percentage of children (Age < 18) who survived
27 children = data[data['Age'] < 18]
28 children_survival_rate = children['Survived'].mean()
29 print(children_survival_rate)
30
31 # 6. Percentage of adults (Age >= 18) who survived
```

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